Assignment: 02

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Section : "A"

Deppt : Computer Science

Subject : Discrete Structure

Q1: Given the following universel set U and its two subsets P and Q, where:

- $U = \{x \mid x \in \mathbb{Z}, 0 \leq x \leq 10\}$
- $P = \{x \mid x \text{ is a prime number}\}$
- $Q = \{x \mid x\,2 < 70\}$

→ (i) Draw a Ven Diagram for the given

→ (ii) List the elements in $P^c \cap Q$.

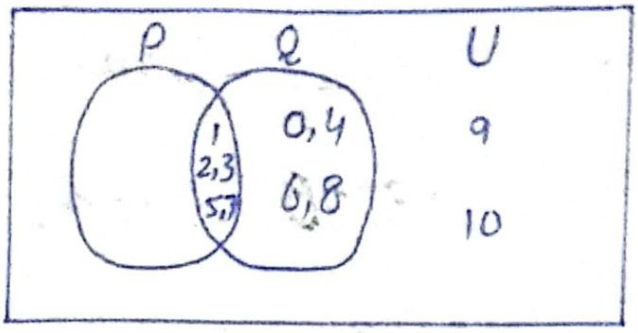
(i) Solution:

$U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$P = \{1, 2, 3, 5, 7\}$

$Q = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

⇒ Ven Diagram:



(ii) List the elements in $P^c \cap Q$.

$P = \{1, 2, 3, 5, 7\}$

$P^c = \{0, 4, 6, 8, 9, 10\}$

$Q = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

$P^c \cap Q = \{0, 4, 6, 8\}$

Q2:- Use Venn Diagram to represent the following.

i) $(A \cap B) \cap C^c$

ii) $A^c \cup (B \cup C)$

iii) $(A - B) \cap C$

iv) $(A \cap B^c) \cup C^c$

⇒ Solution:

(i) $(A \cap B) \cap C^c$

Let;

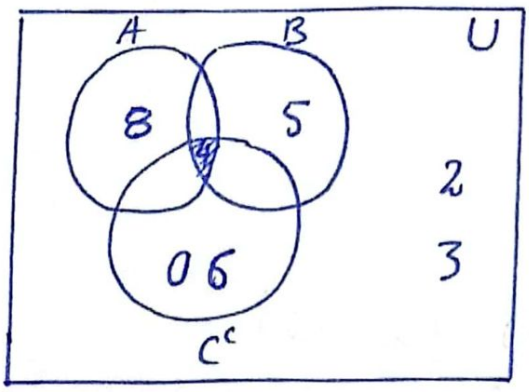
$U = \{0, 2, 3, 4, 6\}$

$A = \{3, 4\}$

$B = \{4, 5\}$

$C = \{2, 8\}$

⇒ Venn Diagram



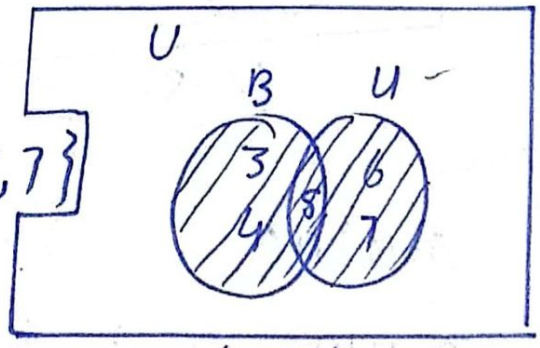
(ii) $A^c \cup (B \cup C)$

Solution

$A^c = \{4, 5, 6, 7\}$

$B \cup C = \{3, 4, 5\} \cup \{5, 6, 7\}$

$= \{3, 4, 5, 6, 7\}$



$(B \cup C)$

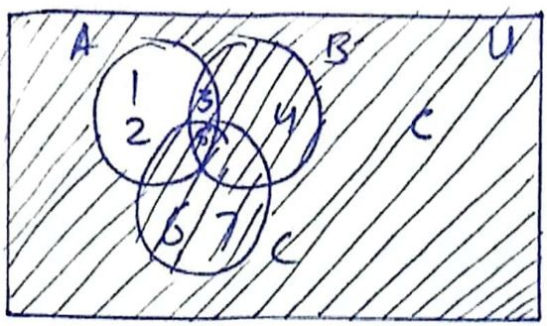
Now

$A^c \cup (B \cup C)$

$\Rightarrow A^c \cup (B \cup C) = \{4, 5, 6, 7\} \cup \{3, 4, 5, 6, 7\}$

$\Rightarrow A^c \cup (B \cup C) = \{3, 4, 5, 6, 7\}$

⇒ Venn Diagram

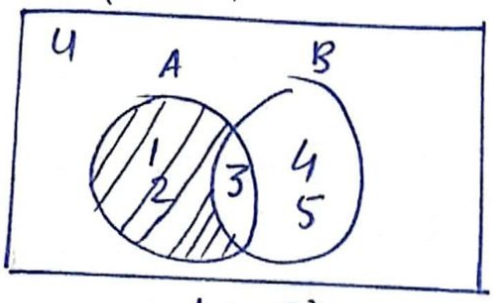


III. $(A-B) \cap C$

Solution

$A-B = \{1,2,3\} - \{3,4,5\}$

$\Rightarrow \{1, 2\}$



$(A-B)$

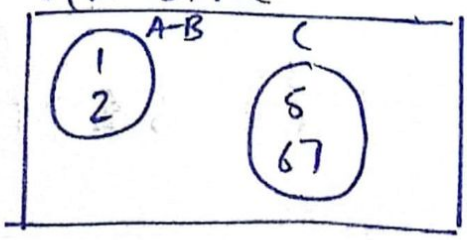
Now

$(A-B) \cap C$

$\Rightarrow (A-B) \cap C = \{1, 2\} \cap \{5,6\}$

$\Rightarrow \{\ \}$

$\therefore (A-B) \cap C$



iv) $(A \cap B^c) \cup C^c$

$A = \{3, 4\}$

$B = \{4, 5\}$

$B^c = \{1, 2, 3\}$

$C = \{2, 3\}$

$C^c = \{1, 4, 5\}$

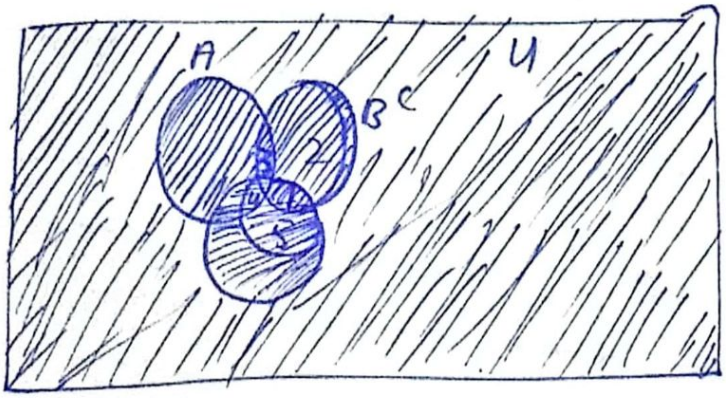
First Find $(A \cap B^c)$

$A \cap B^c = \{3, 4\} \cap \{1, 2, 3\}$

$= \{3\}$

Now find $(A \cap B^c) \cup C^c$

$(A \cap B^c) \cup C^c = \{3\} \cup \{1, 4, 5\}$

$= \{1, 3, 4, 5\}$

Venn Diagram: -